

Failure to Thrive

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Failure to Thrive (FTT)

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Outline

Definition · Aetiology (organic, non-organic, mixed) · Mechanisms of organic FTT · Psychosocial basis of non-organic FTT · Diagnosis & the growth chart · Which parameter falls first · Investigations · Treatment · Prognosis

Definition

Failure to thrive is:

- **Weight consistently below the 3rd-5th percentile** for age and sex, OR
- **A progressive decrease in weight to below the 3rd-5th percentile**, OR
- **A decrease across 2 major growth percentiles** in a short period.

Aetiology

The physiologic basis of FTT of **any** cause is **inadequate nutrition**. It is divided into **organic, non-organic and mixed** – and **most cases are mixed**.

Organic FTT – growth failure due to an acute or chronic disorder that interferes with **nutrient intake, absorption, metabolism or excretion**, or that **increases energy requirements**.

Mechanism	Example disorders
↓ Nutrient intake	cleft lip/palate, CNS disease (cerebral palsy), GERD, parasites, pyloric stenosis, rumination
Malabsorption	coeliac disease, cystic fibrosis, disaccharidase (lactase) deficiency, IBD, short gut
Impaired metabolism	chromosomal (Down, Turner), fructose intolerance, classic galactosaemia, inborn errors of metabolism
↑ Excretion	diabetes mellitus, proteinuria
↑ Energy requirement	bronchopulmonary dysplasia, cystic fibrosis, heart failure, hyperthyroidism, infection

Non-organic FTT – due to **insufficient calorie intake**; usually manifests as **failure to gain weight**, with length and head circumference remaining normal for a time until they too are affected. This is the **most common** pattern.

Up to 80% of children with growth failure have no organic disorder – growth fails because of **environmental neglect (lack of food), stimulus deprivation, or both**.

Psychosocial FTT – growth failure accompanies or precedes poor weight gain, thought to arise because **mental stress raises counter-regulatory hormones (corticosteroids, catecholamines)** that oppose growth hormone.

Lack of food may be due to **poverty, poor feeding technique, over-diluted formula (to stretch it), or an inadequate breast-milk supply** (stressed, exhausted or poorly nourished mother). Non-organic FTT is often a **disordered child-caregiver interaction**; the psychologic basis can resemble **hospitalism** (depression from stimulus deprivation -> apathetic, anorexic child).

The caregiver may be **depressed/apathetic, have poor parenting skills, be anxious or unfulfilled by the role, feel hostile toward the child, or be responding to external stresses** (other children, marital dysfunction, loss, financial difficulty). But poor caregiving does not explain every case – **the child's temperament shapes nurturance too** (a difficult feeder can uncover a parent-child mismatch).

Mixed FTT – organic and non-organic causes overlap: children with organic disease may also have disturbed environments, and severe non-organic undernutrition can produce organic problems.

Diagnosis

Requires a **thorough medical, family and social history, a diet history, laboratory testing** and **frequent weight monitoring**.

- Organic FTT can present at **any age**; most non-organic FTT shows growth failure **before age 1 year**, many by **6 months**.
- Plot age against **weight, height and head size** on WHO charts (0-2 yr) and CDC charts (≥ 2 yr). **Correct for gestation until 2 years** in premature infants.

Weight is the most sensitive indicator of nutritional status.

Which parameter falls first tells you the cause:

- Inadequate intake -> **weight falls from its percentile before length**.

- Reduced **linear growth (height)** = severe, prolonged undernutrition.
- **Simultaneous** fall of length and weight -> suggests a **primary growth disorder or prolonged inflammatory state** (organic).
- **Head circumference falls late** (the brain is preferentially spared) -> indicates **very severe or long-standing** undernutrition.

Underweight children may be **smaller/shorter**, with **fussiness, lethargy/sleepiness, constipation**, and **physical (sitting, walking), social and (in older children) pubertal delays**.

No single test distinguishes organic from non-organic FTT. Because both can coexist, search **simultaneously** for a physical problem and psychosocial factors. Evaluation is ideally **multidisciplinary** (physician, nurse, social worker, nutritionist, child-development expert, ± psychiatrist/psychologist), and **feeding behaviour must be observed** with both staff and parents.

Engage parents as co-investigators – it protects self-esteem and avoids blaming already-guilty parents. Report suspected neglect/abuse to social services, but **referral for preventive support/education is often more appropriate**. Hospitalised children who **gain weight with proper feeding** are more likely to have **non-organic FTT** (though some non-organic children paradoxically lose weight in hospital).

Investigations

Extensive testing is usually unproductive. If history/examination don't point to a cause, most experts **limit screening** to:

- **FBC with differential; ESR**
- **Urea, creatinine, electrolytes**
- **Urinalysis** (including concentrating/acidifying ability) **and culture**
- **Stool** for pH, reducing substances, odour, colour, consistency, fat content
- Depending on community prevalence: **blood lead, HIV, or TB testing**

Sometimes appropriate: **T4 and TSH; growth-hormone assessment** (fasting **IGF-1** and **IGFBP-3**) when height is more affected than weight or both fall together; a **sweat test** for cystic fibrosis if recurrent respiratory disease, ravenous appetite, foul bulky stools, hepatomegaly or family history; review **newborn screening** results. Reserve infection work-up for signs of infection (though a **urine culture** helps as some UTIs lack other signs); reserve radiology for anatomic/functional pathology; check **bone age** if an endocrine cause is suspected.

Treatment

Three aims: **sufficient nutrition, treatment of the underlying disorder, and long-term social support**.

- Provide a nutritious diet with adequate calories for **catch-up growth – about 150% of the normal requirement**.
- **All children grow when given sufficient nutrition** – so weight gain in hospital does not by itself prove non-organic FTT.
- Organic/mixed FTT -> **treat the underlying disorder promptly**. Non-organic/mixed FTT -> **education and emotional support** to repair the parent-child relationship, with **referral to community agencies** (a written predischarge plan with clear accountability is ideal; invite parents to a summary session).

- Some cases need **foster care**; return to biologic parents should be based on **demonstrated ability to care**, not merely the passage of time.

Prognosis

- **Organic FTT** – depends on the cause.
- **Non-organic FTT** – most children **> 1 year** achieve a stable weight above the 3rd percentile.
- Children who develop FTT **before age 1** are at high risk of **cognitive delay** (verbal and maths), **highest** if diagnosed **< 6 months** (peak postnatal brain growth).
- **General behavioural problems in ~50%**; eating/elimination problems in a similar proportion.

References

Lecture material of Dr Ijeoma Stella; MSD/Merck Manual; standard paediatric texts.